

Deviance

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EDMs and GLMs

We say that a distribution belongs to the **exponential dispersion model** family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

For iid observations $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$, a **generalized linear model** (GLM) consists of two components:

$$\begin{aligned} Y_i | \mathbf{x}_i &\sim \text{EDM}(\mu_i, \phi) \\ g(\mu_i) &= \mathbf{x}_i^T \boldsymbol{\beta} \end{aligned}$$

Notice: We want to model the mean μ of an EDM, but the probability function is written in terms of θ .

Rewriting the probability function

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

Consider $y\theta - \kappa(\theta)$ (piece involving θ)

Let $t(y, \mu) = y\theta - \kappa(\theta)$ (θ, μ are 1-to-1 functions of each other)

Estimation: choose μ to maximize likelihood, i.e.
maximize $y\theta - \kappa(\theta)$

$$\frac{d}{d\theta} t(y, \mu) = y - \frac{d\kappa(\theta)}{d\theta} = y - \mu$$

$$\frac{d^2}{d\theta^2} t(y, \mu) = -\frac{d\mu}{d\theta} = -v(\mu) < 0$$

So: $t(y, \mu)$ has a unique maximum at $\mu = y$
i.e. maximizing likelihood $\Leftrightarrow$ make μ close to y

Unit deviance and dispersion model form

$$t(y, \mu) = y\theta - K(\theta)$$

Unit deviance: $d(y, \mu) = 2[t(y, y) - t(y, \mu)]$

$t(y, \mu)$ has a unique maximum at $\mu = y$
 $\Rightarrow t(y, y) \geq t(y, \mu) \quad \forall \mu$

$\Rightarrow d(y, \mu) \geq 0$, and $d(y, \mu) = 0$ iff $\mu = y$

$d(y, \mu)$ measures "distance" between y & μ .

$$f(y; \theta, \phi) = a(y, \phi) \exp\left\{ \frac{y\theta - K(\theta)}{\phi} \right\} = a(y, \phi) \exp\left\{ \frac{t(y, \mu)}{\phi} \right\}$$

$$= a(y, \phi) \exp\left\{ \frac{t(y, y) - t(y, y) + t(y, \mu)}{\phi} \right\}$$

$$= b(y, \phi) \exp\left\{ - \frac{(t(y, y) - t(y, \mu))}{\phi} \right\}$$

$$= b(y, \phi) \exp\left\{ - \frac{d(y, \mu)}{2\phi} \right\}$$

Dispersion model form!

Example: Normal

Previously: $\theta = \mu$ $K(\theta) = \frac{\mu^2}{2}$ $\phi = \sigma^2$

$$t(y, \mu) = y\mu - \frac{\mu^2}{2}$$

$$\begin{aligned}\Rightarrow d(y, \mu) &= 2(t(y, y) - t(y, \mu)) \\ &= 2\left(y^2 - \frac{y^2}{2} - y\mu + \frac{\mu^2}{2}\right) \\ &= \boxed{(y - \mu)^2}\end{aligned}$$

$$N(\mu, \sigma^2) : \quad f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y - \mu)^2}{2\sigma^2}\right\}$$

Dispersion model form!

Example: Bernoulli

$$Y \sim \text{Bernoulli}(\mu)$$

$$\phi = 1$$

$$\theta = \log\left(\frac{\mu}{1-\mu}\right)$$

$$\kappa(\theta) = -\log(1-\mu)$$

$$t(y, \mu) = y \log\left(\frac{\mu}{1-\mu}\right) + \log(1-\mu)$$

$$= y \log(\mu) + (1-y) \log(1-\mu)$$

$$t(y, y) = y \log(y) + (1-y) \log(1-y)$$

Convention: $0 \log(0) = 0$

$$d(y, \mu) = 2 \left(y \log(y) - y \log(\mu) + (1-y) \log(1-y) - (1-y) \log(1-\mu) \right)$$

$$= 2 \left(y \log\left(\frac{y}{\mu}\right) + (1-y) \log\left(\frac{1-y}{1-\mu}\right) \right)$$

if $y=0$: $d(0, \mu) = 2 \log\left(\frac{1}{1-\mu}\right) \xrightarrow[\text{as } \mu \rightarrow 0]{0}$

$y=1$: $d(1, \mu) = 2 \log\left(\frac{1}{\mu}\right) \xrightarrow[\text{as } \mu \rightarrow 1]{0}$

Why do we care?

$$y_i | x_i \sim \text{EDM}(\mu_i, \sigma) \quad g(\mu_i) = x_i^T \beta$$

$$f(y; \mu, \sigma) = b(y, \sigma) \exp \left\{ -\frac{d(y, \mu)}{2\sigma} \right\}$$

Suppose σ is known

$$\text{LRT: } \Lambda = \frac{L_{\text{reduced}}}{L_{\text{full}}} \quad -2 \log \Lambda \approx \chi^2$$

$$L_{\text{reduced}} \propto \exp \left\{ -\frac{1}{2\sigma} \sum_{i=1}^n d(y_i, \hat{\mu}_{\text{reduced}}) \right\}$$

$$L_{\text{full}} \propto \exp \left\{ -\frac{1}{2\sigma} \sum_{i=1}^n d(y_i, \hat{\mu}_{\text{full}}) \right\}$$

$$-2 \log \Lambda = \frac{1}{\sigma} \left[\underbrace{\sum_{i=1}^n d(y_i, \hat{\mu}_{\text{reduced}})}_{\text{residual deviance}} - \underbrace{\sum_{i=1}^n d(y_i, \hat{\mu}_{\text{full}})}_{\text{residual deviance}} \right]$$

Activity

Work on the activity (handout).